



Errors, Convergence & Stopping Criteria

The Big Picture Intuition

If a carpenter makes a 1 cm measuring mistake while building a dining table, it's barely noticeable. But if an aerospace engineer makes a 1 cm mistake on a jet turbine blade, the engine fails catastrophically!

In numerical methods, we rarely get exact answers; we get **iterative approximations**. Quantifying **errors**, understanding how fast algorithms reach the answer (**order of convergence**), and knowing **when to stop computing** are the core pillars of numerical computing.

1. Quantification of Errors

Let x_{true} represent the exact theoretical value, and x_{approx} represent the computed numerical approximation.

A. Absolute Error (E_a)

The raw distance between the true value and your approximation:

$$E_a = |x_{\text{true}} - x_{\text{approx}}|$$

- *Unit*: Same physical units as the variable x .
- *Limitation*: Does not tell you how significant the error is relative to the scale of the object.

B. Relative Error (E_r)

The fraction of the true value that is incorrect (scale-free and dimensionless):

$$E_r = \frac{|x_{\text{true}} - x_{\text{approx}}|}{|x_{\text{true}}|}, \quad x_{\text{true}} \neq 0$$

C. Percentage Error (E_p)

Relative error scaled into a percentage:

$$E_p = E_r \times 100\% = \frac{|x_{\text{true}} - x_{\text{approx}}|}{|x_{\text{true}}|} \times 100\%$$

2. Approximate Relative Error in Iterative Algorithms (ϵ_a)

In real-world engineering, **we don't know** x_{true} (if we knew it, we wouldn't be running an iterative solver!). Therefore, we compare the newest guess x_{k+1} against the previous guess x_k :

Iterative Relative Error Formula

$$\epsilon_a = \left| \frac{x_{k+1} - x_k}{x_{k+1}} \right| \times 100\%, \quad x_{k+1} \neq 0$$

Iterations are halted once $\epsilon_a < \epsilon_s$, where ϵ_s is the pre-specified tolerance threshold.

Significant Digits Criterion (Scarborough's Criterion)

To guarantee that a numerical result is accurate to at least m **significant digits**, iterations must satisfy:

$$\epsilon_a \leq 0.5 \times 10^{2-m}\%$$

Example: For $m = 3$ significant digits, set $\epsilon_s = 0.5 \times 10^{2-3}\% = 0.5 \times 10^{-1}\% = 0.05\%$.

3. Order of Convergence (p) & Error Growth

Let an algorithm generate approximations $x_0, x_1, x_2, \dots$ converging to the true root α . Define the error at iteration step k as:

$$e_k = |x_k - \alpha|$$

If there exist positive real constants $p \geq 1$ and $C > 0$ such that:

$$\lim_{k \rightarrow \infty} \frac{e_{k+1}}{e_k^p} = C$$

then:

- p is the **Order of Convergence**.
- C is the **Asymptotic Error Constant** (if $p = 1$, we must have $C < 1$ for convergence).

Linear ($p=1$):	$e_{k+1} \approx C \cdot e_k$	(Error shrinks by constant fraction)
Superlinear:	$e_{k+1} \approx C \cdot e_k^{1.618}$	(Error shrinks faster and faster)

Quadratic (p=2): $e_{k+1} \approx C \cdot e_k^2$ (Correct decimal digits DOUBLE each step!)

Convergence Speed Classifications

Order (p)	Classification	Error Behavior (e_{k+1})	Typical Digits Behavior	Representative Algorithms
$p = 1$	Linear	$e_{k+1} \approx Ce_k$ ($C < 1$)	Adds a constant fraction of a digit per step	Bisection Method ($C = 0.5$), Fixed-Point Iteration
$p \approx 1.618$	Superlinear	$e_{k+1} \approx Ce_k^{1.618}$	Golden ratio speedup	Secant Method
$p = 2$	Quadratic	$e_{k+1} \approx Ce_k^2$	Correct digits DOUBLE every iteration	Newton-Raphson Method (simple roots)
$p = 3$	Cubic	$e_{k+1} \approx Ce_k^3$	Correct digits TRIPLE every iteration	Halley's Method

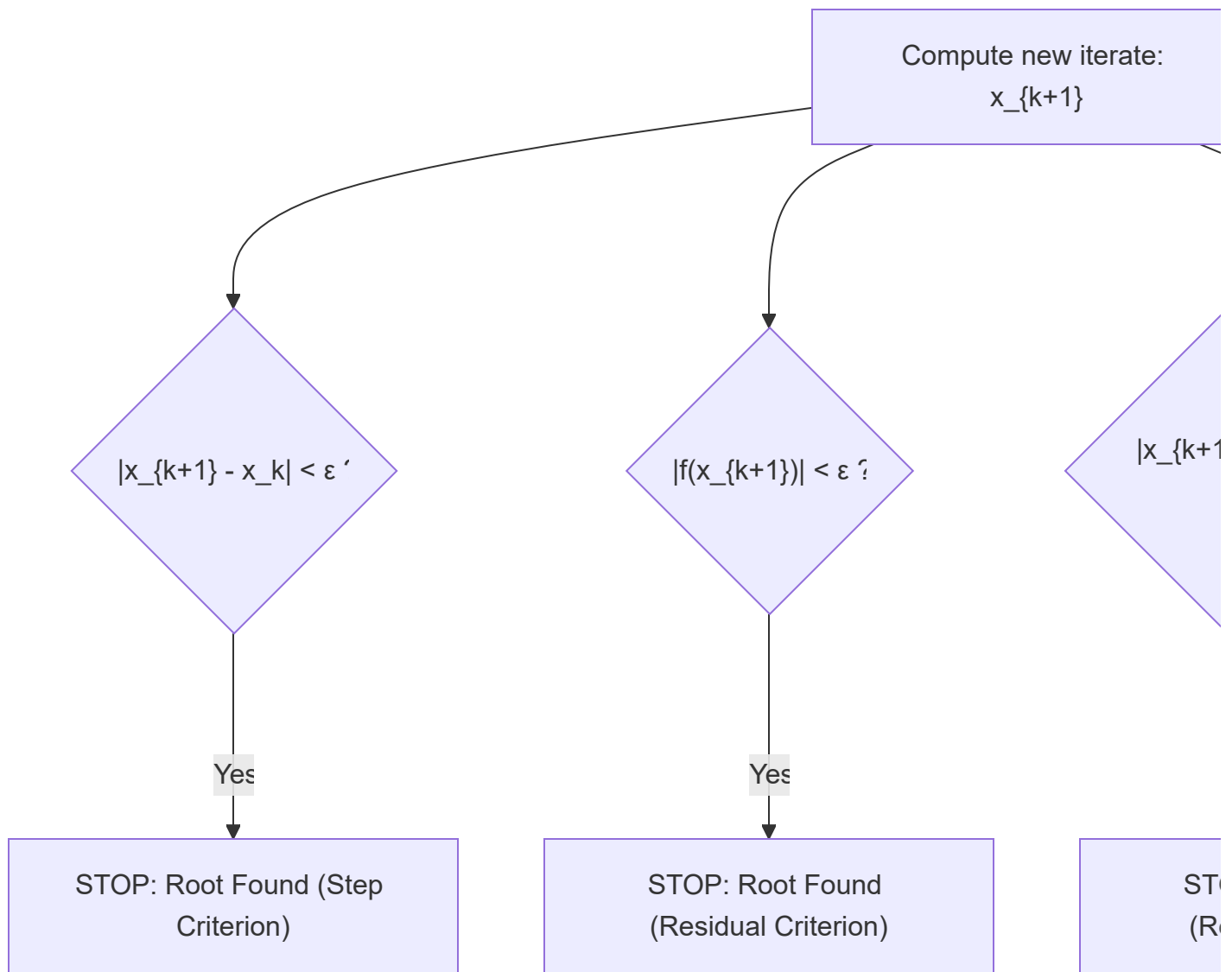
The "Digit Doubling" Superpower of Quadratic Convergence

If an algorithm has quadratic convergence ($p = 2$) with $C \approx 1$:

- Step 1: Error $\approx 10^{-1}$ (1 accurate decimal place)
- Step 2: Error $\approx (10^{-1})^2 = 10^{-2}$ (2 accurate decimal places)
- Step 3: Error $\approx (10^{-2})^2 = 10^{-4}$ (4 accurate decimal places)
- Step 4: Error $\approx (10^{-4})^2 = 10^{-8}$ (8 accurate decimal places)
- Step 5: Error $\approx (10^{-8})^2 = 10^{-16}$ (Machine precision reached in just 5 steps!)

4. Stopping Criteria

Iterative algorithms must terminate cleanly using well-defined stopping rules:



1. Step Difference Criterion:

$$|x_{k+1} - x_k| < \epsilon$$

2. Residual Criterion (Function Value Near Zero):

$$|f(x_{k+1})| < \epsilon$$

3. Relative Step Difference Criterion (Scale-Invariant & Most Robust):

$$\frac{|x_{k+1} - x_k|}{|x_{k+1}|} < \epsilon, \quad x_{k+1} \neq 0$$

4. Safety Counter:

$$k \geq N_{\max} \quad (\text{e.g., 100 iterations to prevent infinite loops})$$

5. Worked Step-by-Step Examples

Worked Example 1: Full Error Analysis on Approximations

Problem

The true value of $\sqrt{2}$ is $x_{\text{true}} \approx 1.41421356$. A numerical method produces consecutive approximations:

- $x_1 = 1.500000$
- $x_2 = 1.416667$

Calculate for iteration 2:

1. Absolute Error E_a
2. Relative Error E_r
3. Percentage Error E_p
4. Approximate Relative Error ϵ_a

Step 1: Compute Absolute Error (E_a)

$$E_a = |x_{\text{true}} - x_2| = |1.41421356 - 1.416667| = |-0.00245344| = \mathbf{0.00245344}$$

Step 2: Compute Relative Error (E_r)

$$E_r = \frac{E_a}{|x_{\text{true}}|} = \frac{0.00245344}{1.41421356} \approx \mathbf{0.00173484}$$

Step 3: Compute Percentage Error (E_p)

$$E_p = E_r \times 100\% = 0.00173484 \times 100\% = \mathbf{0.1735\%}$$

Step 4: Compute Approximate Relative Error (ϵ_a)

Comparing x_2 with x_1 without knowing x_{true} :

$$\epsilon_a = \left| \frac{x_2 - x_1}{x_2} \right| \times 100\% = \left| \frac{1.416667 - 1.500000}{1.416667} \right| \times 100\% = \left| \frac{-0.083333}{1.416667} \right| \times 100\% = \mathbf{5.882\%}$$

Worked Example 2: Determining Convergence Order from Numerical Data

Problem

An unknown iterative algorithm produces the following error sequence e_k across 4 iterations:

- $e_0 = 0.20000$
- $e_1 = 0.04000$
- $e_2 = 0.00160$
- $e_3 = 0.00000256$

Determine the order of convergence p and the asymptotic error constant C .

Step 1: Form the error ratios for $p = 1$ (Linear Hypothesis)

$$\frac{e_1}{e_0} = \frac{0.04}{0.20} = 0.20$$

$$\frac{e_2}{e_1} = \frac{0.0016}{0.04} = 0.04 \neq 0.20 \implies \text{Not linear!}$$

Step 2: Form the error ratios for $p = 2$ (Quadratic Hypothesis)

$$\frac{e_1}{e_0^2} = \frac{0.04}{(0.20)^2} = \frac{0.04}{0.04} = 1.00$$

$$\frac{e_2}{e_1^2} = \frac{0.0016}{(0.04)^2} = \frac{0.0016}{0.0016} = 1.00$$

$$\frac{e_3}{e_2^2} = \frac{0.00000256}{(0.0016)^2} = \frac{2.56 \times 10^{-6}}{2.56 \times 10^{-6}} = 1.00$$

Step 3: Conclusion

- Since $\frac{e_{k+1}}{e_k^2} = 1.00 = \text{constant}$ for all k :
- **Order of Convergence:** $p = 2$ (Quadratic Convergence).
- **Asymptotic Error Constant:** $C = 1.00$.

6. Active Recall & Exam Self-Test

? Practice Question: Scarborough Criterion

If an engineer requires an approximate root to be correct to at least 4 significant figures, what is the maximum permissible approximate relative error ϵ_s ?

✓ Step-by-Step Solution >

1. Use Scarborough's formula with $m = 4$:

$$\epsilon_s = 0.5 \times 10^{2-m}\%$$

2. Substitute $m = 4$:

$$\epsilon_s = 0.5 \times 10^{2-4}\% = 0.5 \times 10^{-2}\% = \mathbf{0.005\%}$$

3. The algorithm can safely stop when $\epsilon_a \leq 0.005\%$.

Related Notes

- [Roots of Equations](#) — Mathematical zeros and existence proofs.
- [Bisection Method](#) — Linear rate ($p = 1$, $C = 0.5$).
- [Newton-Raphson Method](#) — Benchmark quadratic solver ($p = 2$).
- [Secant Method](#) — Superlinear solver ($p \approx 1.618$).